

pyROX: Rapid Opacity X-sections

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Summary

In recent years, significant advances have been made in exoplanet and brown dwarf observations. By using state-of-the-art models, astronomers can determine properties of their atmospheres, such as temperatures, the presence of clouds, or the chemical abundances of molecules and atoms. Accurate and up-to-date opacities, which describe how strongly atmospheric gases absorb radiation, are therefore crucial to avoid inconclusive or biased results. Due to continuous expansion of the line lists that tabulate the quantum transitions, however, it has become challenging to compute the wavelength-dependent cross sections of the diverse array of molecules and atoms.

We introduce pyROX, an easy-to-use Python package to calculate molecular and atomic cross sections. Since pyROX works on CPUs, it can compute a small line list on a regular workstation, but it is also easily parallelised on a cluster for larger line lists. In addition to line opacities, pyROX supports calculations of collision-induced absorption. Tutorials are provided in the online documentation which explain the configuration parameters and different functionalities of pyROX.

Statement of need

The advent of a new generation of telescopes and instruments has led to a dramatically increased quality in observations of exoplanets and brown dwarfs. Such sub-stellar objects are now observed over a wide wavelength range (1–20 μm) with JWST spectra (e.g. [August et al., 2023](#); [Carter et al., 2024](#); [Matthews et al., 2025](#); [Miles et al., 2023](#)), for instance, which was previously difficult to access. Developments in ground-based instrumentation allow astronomers to measure young exoplanet companions at closer separations to their host stars (e.g. [Landman et al., 2024](#); [Xuan, Mérand, et al., 2024](#)) and at high spectral resolutions (e.g. [Nortmann et al., 2025](#); [Xuan, Hsu, et al., 2024](#)). At the same time, progress has also been made in atmospheric modelling using software for radiative transfer, chemistry, circulation models, etc. (e.g. [Mollière et al., 2019](#); [Stock et al., 2018](#); [Wardenier et al., 2021](#)). Recently, these observations and software are coupled with sampling algorithms to characterise the atmospheres of the sub-stellar objects (e.g. [Barrado et al., 2023](#); [Brogi & Line, 2019](#); [Gibson et al., 2020](#); [Line et al., 2015](#)).

Opacity cross sections play a key role in accurately modelling sub-stellar atmospheres. Opacity governs the dominant energy transport mechanism (i.e. radiation or convection), which affects the thermal structure of the atmosphere ([Marley et al., 2021](#)). Furthermore, high-resolution studies require well-determined frequencies for the transition lines. Inaccuracies in line-list data can result in biased abundance constraints (e.g. [Brogi & Line, 2019](#); [de Regt et al., 2024](#); [Niraula et al., 2022, 2023](#)) or ambiguous (non)-detections of certain molecular gases (e.g. [de Regt et al., 2022](#); [Merritt et al., 2020](#); [Serindag et al., 2021](#); [Simonnin et al., 2025](#)). It is

therefore important that the most up-to-date and complete opacity data are used. However, it can be difficult to efficiently calculate opacity cross sections from line lists that sometimes consist of billions of transitions.

To help resolve this challenge, we present pyROX, a user-friendly Python package to calculate molecular and atomic cross sections for applications in models of sub-stellar atmospheres. pyROX supports line opacity calculations from the ExoMol ([Tennyson et al., 2024](#)), HITRAN ([Gordon et al., 2022](#)), HITEMP ([Rothman et al., 2010](#)), and [Kurucz databases](#). Collision-induced absorption (CIA) coefficients can also be calculated from the HITRAN and [Borysow](#) databases.

State of the field

Existing open-source codes, such as [Cthulhu](#) ([Agrawal & MacDonald, 2024](#)), [ExoCross](#) ([Yurchenko et al., 2018](#); [J. Zhang et al., 2024](#)) and [HELIOS-K](#) ([Grimm et al., 2021](#); [Grimm & Heng, 2015](#)), can calculate cross sections at comparable performance to pyROX. However, ExoCross is written in Fortran and HELIOS-K utilises GPU acceleration, which can limit their use to experts with the appropriate hardware. pyROX is a Python code that runs only on CPUs, which should make it accessible for the opacity needs and hardware of most astronomers. In contrast to Cthulhu, pyROX supports cross-section calculations on any user-provided wavelength or wavenumber grid. This enables the user to fix the spectral resolution ($\mathcal{R} = \lambda/\Delta\lambda$), which cannot be achieved with equal spacing in wavelength or wavenumber. Unlike the aforementioned codes, pyROX can also account for collision-induced absorption, which is a considerable source of opacity in sub-stellar atmospheres ([Freedman et al., 2014](#)).

Software design

We aim to provide an easy-to-use and comprehensive tool to calculate cross sections for atmospheric gases. Although more efficient languages exist, we chose to develop pyROX in Python to keep it readily accessible to the astronomy community. The main workflow of pyROX is as follows:

- **Download and read files:** The necessary input files (line lists, partition functions, broadening coefficients or CIA files) can be downloaded with a simple command. When reading the relevant parameters, pyROX automatically handles the different data structures of the supported databases (ExoMol, HITRAN/HITEMP, Kurucz, Borysow).
- **Compute line strengths and widths:** For line-opacity calculations, pyROX calculates the strength and broadening widths for each line transition at the user-provided pressure and temperature. Support is offered for various [pressure-broadening descriptions](#), namely (1) transition-specific broadening from the ExoMol data, (2) parameterised broadening based on the rotational quantum numbers, or (3) constant coefficients. At this point, pyROX can speed up the line-profile computation by selecting only the main line-strength contributors.
- **Compute line profiles:** Next, pyROX computes the Voigt profiles as the real part of the Faddeeva function (Eq. 12 of [Gandhi et al., 2020](#)). Although faster approximations exist ([Schreier, 2018](#)), we currently support only the more accurate [scipy.special.wofz implementation](#). On a single CPU, this computes 10000 lines in about 3 seconds, using $\Delta\nu = 0.01 \text{ cm}^{-1}$ and a wing-cutoff of 100 cm^{-1} .
- **Combine cross sections:** The line profiles are summed into wavelength-dependent cross sections for each temperature-pressure point. Multiple temperature-pressure points can be calculated simultaneously through the use of the [joblib](#) package, enabling straightforward CPU parallelisation for larger line lists.
- **Format and save:** The cross-section table is saved into an efficient HDF5 output file. For CIA calculations, pyROX restructures the coefficients read from the input files into a wavelength- and temperature-dependent grid and also saves these data to an HDF5 file.

pyROX offers built-in support for converting its output into the high-resolution opacities used by petitRADTRANS (Mollière et al., 2019). In future releases, we plan to add conversions for other radiative transfer codes popular in the exoplanet and brown dwarf communities.

The [documentation for pyROX](#) includes examples to run the code and the detailed tutorials allow new users to get up to speed with the available options and functionalities. We also welcome suggestions for new features, which can be done by [opening an issue](#) on GitHub. If you want to contribute to pyROX, please read the [documented guidelines](#).

Research impact statement

Recent publications from our team have employed pyROX-computed cross sections (e.g. [de Regt et al., 2025](#); [González Picos, de Regt, et al., 2025](#); [González Picos, Snellen, et al., 2025](#); [Siebenaler et al., 2025](#); [Siebenaler & Miguel, 2026](#)), and this reach is currently expanding to other research groups as well (e.g. [van Sluijs et al., 2025](#); [Y. Zhang et al., 2026](#)).

AI usage disclosure

Generative AI tools (GPT-4o) were used in the writing of docstrings and code refactoring. All AI-generated additions or changes were reviewed and edited by the authors to verify that the intended functionality was maintained. No AI tools were used in the writing of this paper and online documentation.

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References

- Agrawal, A., & MacDonald, R. (2024). Cthulhu: An open source molecular and atomic cross section computation code for substellar atmospheres. *The Journal of Open Source Software*, 9(102), 6894. <https://doi.org/10.21105/joss.06894>
- August, P. C., Bean, J. L., Zhang, M., Lunine, J., Xue, Q., Line, M., & Smith, P. C. B. (2023). Confirmation of subsolar metallicity for WASP-77Ab from JWST thermal emission spectroscopy. *The Astrophysical Journal Letters*, 953(2), L24. <https://doi.org/10.3847/2041-8213/ace828>
- Barrado, D., Mollière, P., Patapis, P., Min, M., Tremblin, P., Ardevol Martinez, F., Whiteford, N., Vasist, M., Argyriou, I., Samland, M., Lagage, P.-O., Decin, L., Waters, R., Henning, T., Morales-Calderón, M., Guedel, M., Vandenbussche, B., Absil, O., Baudoz, P., ... Wright, G. (2023). $^{15}\text{NH}_3$ in the atmosphere of a cool brown dwarf. *Nature*, 624(7991), 263–266. <https://doi.org/10.1038/s41586-023-06813-y>
- Brogi, M., & Line, M. R. (2019). Retrieving temperatures and abundances of exoplanet atmospheres with high-resolution cross-correlation spectroscopy. *The Astronomical Journal*, 157(3), 114. <https://doi.org/10.3847/1538-3881/aaffd3>
- Carter, A. L., May, E. M., Espinoza, N., Welbanks, L., Ahrer, E., Alderson, L., Brahm, R., Feinstein, A. D., Grant, D., Line, M., Morello, G., O’Steen, R., Radica, M., Rustamkulov, Z., Stevenson, K. B., Turner, J. D., Alam, M. K., Anderson, D. R., Batalha, N. M., ... Zhang, X. (2024). A benchmark JWST near-infrared spectrum for the exoplanet WASP-39 b. *Nature Astronomy*, 8, 1008–1019. <https://doi.org/10.1038/s41550-024-02292-x>

- de Regt, S., Gandhi, S., Snellen, I. A. G., Zhang, Y., Ginski, C., González Picos, D., Kesseli, A. Y., Landman, R., Mollière, P., Nasedkin, E., Sánchez-López, A., & Stolker, T. (2024). The ESO SupJup Survey. I. Chemical and isotopic characterisation of the late L-dwarf DENIS J0255-4700 with CRIRES⁺. *Astronomy and Astrophysics*, 688, A116. <https://doi.org/10.1051/0004-6361/202348508>
- de Regt, S., Kesseli, A. Y., Snellen, I. A. G., Merritt, S. R., & Chubb, K. L. (2022). A quantitative assessment of the VO line list: Inaccuracies hamper high-resolution VO detections in exoplanet atmospheres. *Astronomy and Astrophysics*, 661, A109. <https://doi.org/10.1051/0004-6361/202142683>
- de Regt, S., Snellen, I. A. G., Allard, N. F., González Picos, D., Gandhi, S., Grassier, N., Landman, R., Mollière, P., Nasedkin, E., Stolker, T., & Zhang, Y. (2025). The ESO SupJup Survey: VII. Clouds and line asymmetries in CRIRES⁺ J-band spectra of the Luhman 16 binary. *Astronomy and Astrophysics*, 696, A225. <https://doi.org/10.1051/0004-6361/202453190>
- Freedman, R. S., Lustig-Yaeger, J., Fortney, J. J., Lupu, R. E., Marley, M. S., & Lodders, K. (2014). Gaseous mean opacities for giant planet and ultracool dwarf atmospheres over a range of metallicities and temperatures. *The Astrophysical Journal Supplement Series*, 214(2), 25. <https://doi.org/10.1088/0067-0049/214/2/25>
- Gandhi, S., Brogi, M., Yurchenko, S. N., Tennyson, J., Coles, P. A., Webb, R. K., Birkby, J. L., Guilluy, G., Hawker, G. A., Madhusudhan, N., Bonomo, A. S., & Sozzetti, A. (2020). Molecular cross-sections for high-resolution spectroscopy of super-Earths, warm Neptunes, and hot Jupiters. *Monthly Notices of the Royal Astronomical Society*, 495(1), 224–237. <https://doi.org/10.1093/mnras/staa981>
- Gibson, N. P., Merritt, S., Nugroho, S. K., Cubillos, P. E., de Mooij, E. J. W., Mikal-Evans, T., Fossati, L., Lothringer, J., Nikolov, N., Sing, D. K., Spake, J. J., Watson, C. A., & Wilson, J. (2020). Detection of Fe I in the atmosphere of the ultra-hot Jupiter WASP-121b, and a new likelihood-based approach for Doppler-resolved spectroscopy. *Monthly Notices of the Royal Astronomical Society*, 493(2), 2215–2228. <https://doi.org/10.1093/mnras/staa228>
- González Picos, D., de Regt, S., Gandhi, S., Grassier, N., & Snellen, I. A. G. (2025). Disentangling disc and atmospheric signatures of young brown dwarfs with JWST/NIRSpec. *Astronomy and Astrophysics*, 703, A65. <https://doi.org/10.1051/0004-6361/202555669>
- González Picos, D., Snellen, I., & de Regt, S. (2025). Chemical evolution imprints in the rare isotopes of nearby M dwarfs. *Nature Astronomy*, 9(11), 1692–1700. <https://doi.org/10.1038/s41550-025-02641-4>
- Gordon, I. E., Rothman, L. S., Hargreaves, R. J., Hashemi, R., Karlovets, E. V., Skinner, F. M., Conway, E. K., Hill, C., Kochanov, R. V., Tan, Y., Wcisło, P., Finenko, A. A., Nelson, K., Bernath, P. F., Birk, M., Boudon, V., Campargue, A., Chance, K. V., Coustenis, A., ... Yurchenko, S. N. (2022). The HITRAN2020 molecular spectroscopic database. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 277, 107949. <https://doi.org/10.1016/j.jqsrt.2021.107949>
- Grimm, S. L., & Heng, K. (2015). HELIOS-K: An ultrafast, open-source opacity calculator for radiative transfer. *The Astrophysical Journal*, 808(2), 182. <https://doi.org/10.1088/0004-637X/808/2/182>
- Grimm, S. L., Malik, M., Kitzmann, D., Guzmán-Mesa, A., Hoeijmakers, H. J., Fisher, C., Mendonça, J. M., Yurchenko, S. N., Tennyson, J., Alesina, F., Buchschacher, N., Burnier, J., Segransan, D., Kurucz, R. L., & Heng, K. (2021). HELIOS-K 2.0 opacity calculator and open-source opacity database for exoplanetary atmospheres. *The Astrophysical Journal Supplement Series*, 253(1), 30. <https://doi.org/10.3847/1538-4365/abd773>
- Landman, R., Stolker, T., Snellen, I. A. G., Costes, J., de Regt, S., Zhang, Y., Gandhi, S.,

- Molliere, P., Kesseli, A., Vigan, A., & Sanchez-López, A. (2024). β Pictoris b through the eyes of the upgraded CRIRES+. Atmospheric composition, spin rotation, and radial velocity. *Astronomy and Astrophysics*, 682, A48. <https://doi.org/10.1051/0004-6361/202347846>
- Line, M. R., Teske, J., Burningham, B., Fortney, J. J., & Marley, M. S. (2015). Uniform atmospheric retrieval analysis of ultracool dwarfs. I. Characterizing benchmarks, Gl 570D and HD 3651B. *The Astrophysical Journal*, 807(2), 183. <https://doi.org/10.1088/0004-637X/807/2/183>
- Marley, M. S., Saumon, D., Visscher, C., Lupu, R., Freedman, R., Morley, C., Fortney, J. J., Seay, C., Smith, A. J. R. W., Teal, D. J., & Wang, R. (2021). The Sonora brown dwarf atmosphere and evolution models. I. Model description and application to cloudless atmospheres in rainout chemical equilibrium. *The Astrophysical Journal*, 920(2), 85. <https://doi.org/10.3847/1538-4357/ac141d>
- Matthews, E. C., Mollière, P., Kühnle, H., Patapis, P., Whiteford, N., Samland, M., Lagage, P.-O., Waters, R., Tsai, S.-M., Zahnle, K., Guedel, M., Henning, T., Vandenbussche, B., Absil, O., Argyriou, I., Barrado, D., Coulais, A., Glauser, A. M., Olofsson, G., ... Östlin, G. (2025). HCN and C₂H₂ in the atmosphere of a T8.5+T9 brown dwarf binary. *The Astrophysical Journal Letters*, 981(2), L31. <https://doi.org/10.3847/2041-8213/adb4ec>
- Merritt, S. R., Gibson, N. P., Nugroho, S. K., de Mooij, E. J. W., Hooton, M. J., Matthews, S. M., McKemmish, L. K., Mikal-Evans, T., Nikolov, N., Sing, D. K., Spake, J. J., & Watson, C. A. (2020). Non-detection of TiO and VO in the atmosphere of WASP-121b using high-resolution spectroscopy. *Astronomy and Astrophysics*, 636, A117. <https://doi.org/10.1051/0004-6361/201937409>
- Miles, B. E., Biller, B. A., Patapis, P., Worthen, K., Rickman, E., Hoch, K. K. W., Skemer, A., Perrin, M. D., Whiteford, N., Chen, C. H., Sargent, B., Mukherjee, S., Morley, C. V., Moran, S. E., Bonnefoy, M., Petrus, S., Carter, A. L., Choquet, E., Hinkley, S., ... Zhang, Z. (2023). The JWST early-release science program for direct observations of exoplanetary systems II: A 1 to 20 μ m spectrum of the planetary-mass companion VHS 1256-1257 b. *The Astrophysical Journal Letters*, 946(1), L6. <https://doi.org/10.3847/2041-8213/acb04a>
- Mollière, P., Wardenier, J. P., van Boekel, R., Henning, Th., Molaverdikhani, K., & Snellen, I. A. G. (2019). petitRADTRANS. A Python radiative transfer package for exoplanet characterization and retrieval. *Astronomy and Astrophysics*, 627, A67. <https://doi.org/10.1051/0004-6361/201935470>
- Niraula, P., de Wit, J., Gordon, I. E., Hargreaves, R. J., & Sousa-Silva, C. (2023). Origin and extent of the opacity challenge for atmospheric retrievals of WASP-39 b. *The Astrophysical Journal Letters*, 950(2), L17. <https://doi.org/10.3847/2041-8213/acd6f8>
- Niraula, P., de Wit, J., Gordon, I. E., Hargreaves, R. J., Sousa-Silva, C., & Kochanov, R. V. (2022). The impending opacity challenge in exoplanet atmospheric characterization. *Nature Astronomy*, 6, 1287–1295. <https://doi.org/10.1038/s41550-022-01773-1>
- Nortmann, L., Lesjak, F., Yan, F., Cont, D., Czesla, S., Lavail, A., Rains, A. D., Nagel, E., Boldt-Christmas, L., Hatzes, A., Reiners, A., Piskunov, N., Kochukhov, O., Heiter, U., Shulyak, D., Rengel, M., & Seemann, U. (2025). CRIRES⁺ transmission spectroscopy of WASP-127 b: Detection of the resolved signatures of a supersonic equatorial jet and cool poles in a hot planet. *Astronomy and Astrophysics*, 693, A213. <https://doi.org/10.1051/0004-6361/202450438>
- Rothman, L. S., Gordon, I. E., Barber, R. J., Dothe, H., Gamache, R. R., Goldman, A., Perevalov, V. I., Tashkun, S. A., & Tennyson, J. (2010). HITEMP, the high-temperature molecular spectroscopic database. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 111, 2139–2150. <https://doi.org/10.1016/j.jqsrt.2010.05.001>
- Schreier, F. (2018). The Voigt and complex error function: Humlíček's rational approximation

- generalized. *Monthly Notices of the Royal Astronomical Society*, 479(3), 3068–3075. <https://doi.org/10.1093/mnras/sty1680>
- Serindag, D. B., Nugroho, S. K., Mollière, P., de Mooij, E. J. W., Gibson, N. P., & Snellen, I. A. G. (2021). Is TiO emission present in the ultra-hot Jupiter WASP-33b? A reassessment using the improved ExoMol TOTO line list. *Astronomy and Astrophysics*, 645, A90. <https://doi.org/10.1051/0004-6361/202039135>
- Siebenaler, L., & Miguel, Y. (2026). Mean opacity tables for probing the interior and atmosphere of giant planets. *Monthly Notices of the Royal Astronomical Society*, 546(3), staf2205. <https://doi.org/10.1093/mnras/staf2205>
- Siebenaler, L., Miguel, Y., de Regt, S., & Guillot, T. (2025). Conditions for radiative zones in the molecular hydrogen envelope of Jupiter and Saturn: The role of alkali metals. *Astronomy and Astrophysics*, 693, A308. <https://doi.org/10.1051/0004-6361/202452860>
- Simonnin, A., Parmentier, V., Wardenier, J. P., Chauvin, G., Chiavassa, A., N'Diaye, M., Tan, X., Heidari, N., Prinoth, B., Bean, J., Hébrard, G., Line, M., Kitzmann, D., Kasper, D., Pelletier, S., Seidel, J. V., Seifhart, A., Benneke, B., Bonfils, X., ... Smith, P. (2025). Time-resolved absorption of six chemical species with MAROON-X points to a strong drag in the ultra-hot Jupiter TOI-1518 b. *Astronomy and Astrophysics*, 698, A314. <https://doi.org/10.1051/0004-6361/202453241>
- Stock, J. W., Kitzmann, D., Patzer, A. B. C., & Sedlmayr, E. (2018). FastChem: A computer program for efficient complex chemical equilibrium calculations in the neutral/ionized gas phase with applications to stellar and planetary atmospheres. *Monthly Notices of the Royal Astronomical Society*, 479(1), 865–874. <https://doi.org/10.1093/mnras/sty1531>
- Tennyson, J., Yurchenko, S. N., Zhang, J., Bowesman, C. A., Brady, R. P., Buldyreva, J., Chubb, K. L., Gamache, R. R., Gorman, M. N., Guest, E. R., Hill, C., Kefala, K., Lynas-Gray, A. E., Mellor, T. M., McKemmish, L. K., Mitev, G. B., Mizus, I. I., Owens, A., Peng, Z., ... Zobov, N. F. (2024). The 2024 release of the ExoMol database: Molecular line lists for exoplanet and other hot atmospheres. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 326, 109083. <https://doi.org/10.1016/j.jqsrt.2024.109083>
- van Sluijs, L., Rauscher, E., Kempton, E. M.-R., Kennedy, T., Malsky, I., Matsunaga, N., Meyer, M., McWilliam, A., Monnier, J. D., Otsubo, S., Sarugaku, Y., & Takeuchi, T. (2025). First results from WINERED: Detection of emission lines from neutral iron and a combined set of trace species on the dayside of WASP-189 b. *The Astronomical Journal*, 170(4), 217. <https://doi.org/10.3847/1538-3881/adfb61>
- Wardenier, J. P., Parmentier, V., Lee, E. K. H., Line, M. R., & Gharib-Nezhad, E. (2021). Decomposing the iron cross-correlation signal of the ultra-hot Jupiter WASP-76b in transmission using 3D Monte Carlo radiative transfer. *Monthly Notices of the Royal Astronomical Society*, 506(1), 1258–1283. <https://doi.org/10.1093/mnras/stab1797>
- Xuan, J. W., Hsu, C.-C., Finnerty, L., Wang, J., Ruffio, J.-B., Zhang, Y., Knutson, H. A., Mawet, D., Mamajek, E. E., Inglis, J., Wallack, N. L., Bryan, M. L., Blake, G. A., Mollière, P., Hejazi, N., Baker, A., Bartos, R., Calvin, B., Cetre, S., ... Horstman, K. (2024). Are these planets or brown dwarfs? Broadly solar compositions from high-resolution atmospheric retrievals of $\sim 10\text{--}30\text{ M}_{\text{Jup}}$ companions. *The Astrophysical Journal*, 970(1), 71. <https://doi.org/10.3847/1538-4357/ad4796>
- Xuan, J. W., Mérand, A., Thompson, W., Zhang, Y., Lacour, S., Blakely, D., Mawet, D., Oppenheimer, R., Kammerer, J., Batygin, K., Sanghi, A., Wang, J., Ruffio, J.-B., Liu, M. C., Knutson, H., Brandner, W., Burgasser, A., Rickman, E., Bowens-Rubin, R., ... Woillez, J. (2024). The cool brown dwarf Gliese 229 B is a close binary. *Nature*, 634(8036), 1070–1074. <https://doi.org/10.1038/s41586-024-08064-x>
- Yurchenko, S. N., Al-Refaie, A. F., & Tennyson, J. (2018). EXOCROSS: a general program

for generating spectra from molecular line lists. *Astronomy and Astrophysics*, 614, A131. <https://doi.org/10.1051/0004-6361/201732531>

Zhang, J., Tennyson, J., & Yurchenko, S. N. (2024). PYEXOCROSS: a Python program for generating spectra and cross-sections from molecular line lists. *RAS Techniques and Instruments*, 3(1), 257–287. <https://doi.org/10.1093/rasti/rzae016>

Zhang, Y., Wardenier, J. P., Householder, A., Komacek, T. D., Kesseli, A. Y., Dai, F., Howard, A. W., Inglis, J., Isaacson, H., Knutson, H. A., Mawet, D., Pino, L., Wallack, N., Xuan, J. W., Carmichael, T. W., Huber, D., Lee, R. A., Saunders, N., Weiss, L., & Zhang, J. (2026). Extreme winds on the emerging dayside of an ultrahot Jupiter. *The Astrophysical Journal Letters*, 997(2), L40. <https://doi.org/10.3847/2041-8213/ae36a1>